

CHAPTER 14 PARTIAL DERIVATIVES

14.1 Functions of Several Variables

- **Functions of Two Variables** A real-valued function of two real variables, is a function f that assigns to each ordered pair (x, y) in some set D of the plane a unique real number $f(x, y)$.

For example,

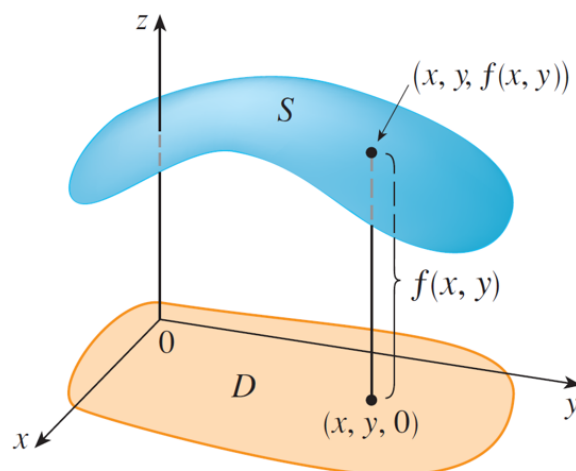
- $f(x, y) = x^2 + 3y^2$

- $g(x, y) = 2x\sqrt{y}$

$$D = \{(x, y) \mid x, y \in \mathbb{R}\}$$

$$D = \{(x, y) \mid x \in \mathbb{R}, y \geq 0\}$$

- The set D is called the **domain** of the function. If it is not specified, we take D to be the natural domain, the set of all (x, y) in the plane for which the function rule makes sense.
- The **range** of a function is its set of values.
- If $z = f(x, y)$, we call x and y **independent variables** and z **dependent variable**.



- **Example** For each of the following functions, evaluate $f(3, 2)$ and find and sketch the domain.

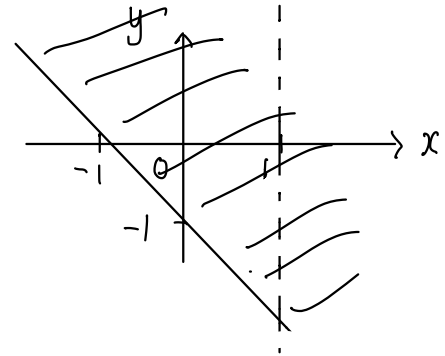
(a) $f(x, y) = \frac{\sqrt{x+y+1}}{x-1}$

$$f(3, 2) = \frac{\sqrt{3+2+1}}{3-1} = \frac{\sqrt{6}}{2}$$

$x=3,$
 $y=2$

$$\begin{cases} \sqrt{x+y+1} \Rightarrow x+y+1 \geq 0 \Rightarrow y \geq -x-1 \\ x-1 \text{ den} \Rightarrow x-1 \neq 0 \Rightarrow x \neq 1 \end{cases}$$

$$D = \{(x, y) \mid y \geq -x-1, x \neq 1\}$$

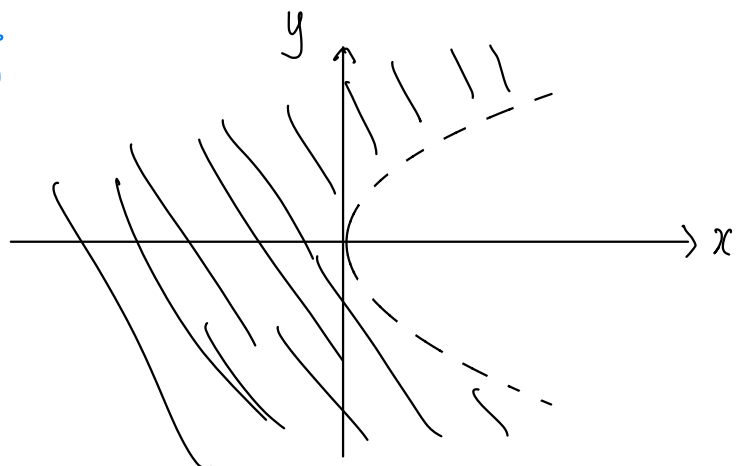


(b) $f(x, y) = x \ln(y^2 - x)$

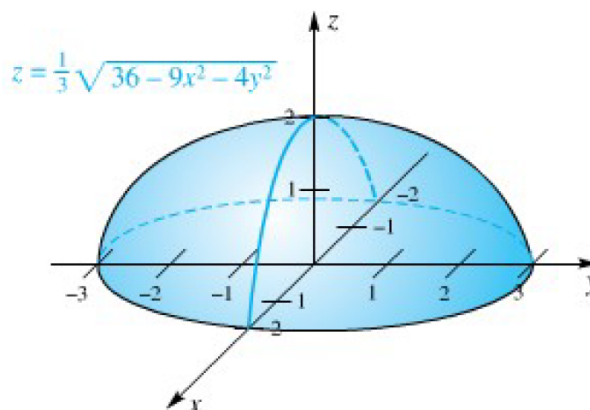
$$f(3, 2) = 3 \ln(2^2 - 3) = 0$$

$$\ln(y^2 - x) \Rightarrow y^2 - x > 0 \Rightarrow x < y^2$$

$$D = \{(x, y) \mid x < y^2\}$$



- **Example** Find the domain and range of $f(x, y) = \frac{1}{3}\sqrt{36 - 9x^2 - 4y^2}$ with the given graph.



$$\begin{aligned} &\downarrow \\ &36 - 9x^2 - 4y^2 \geq 0 \\ &\Rightarrow 9x^2 + 4y^2 \leq 36 \end{aligned}$$

$$D = \{(x, y) \mid 9x^2 + 4y^2 \leq 36\}$$

$$R = \{z \mid 0 \leq z \leq 2\}$$

- **Example** Find the domain and range of $f(x, y) = \sqrt{9 - x^2 - y^2}$.

Domain : $\sqrt{9 - x^2 - y^2} \Rightarrow 9 - x^2 - y^2 \geq 0 \Rightarrow x^2 + y^2 \leq 9$

$$\therefore D = \{(x, y) \mid x^2 + y^2 \leq 9\}$$

Range : $0 \leq x^2 + y^2 \leq 9$
 $0 \geq -x^2 - y^2 \geq -9$

$$9 \geq 9 - x^2 - y^2 \geq 0$$

$$3 \geq \sqrt{9 - x^2 - y^2} \geq 0$$

$$\therefore R = \{z \mid 0 \leq z \leq 3\}$$

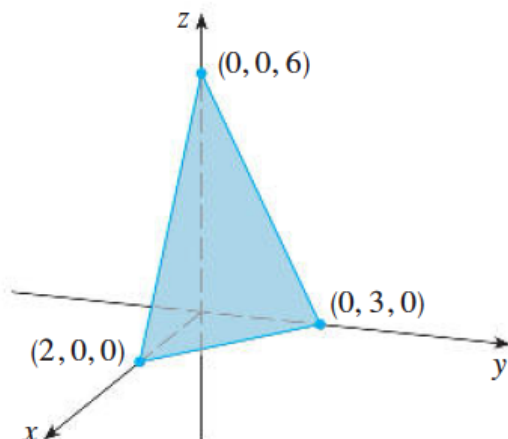
- **Example** How to sketch the graph of $f(x, y) = 6 - 3x - 2y$.

First, notice that the graph is a plane.

$$x\text{-intercept} : y = z = 0 \Rightarrow x = 2$$

$$y\text{-intercept} : x = z = 0 \Rightarrow y = 3$$

$$z\text{-intercept} : x = y = 0 \Rightarrow z = 6$$



Linear Function: $f(x, y) = ax + by + c$

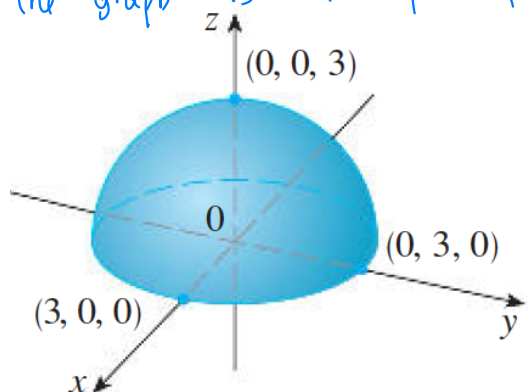
- **Example** How to sketch the graph of $f(x, y) = \sqrt{9 - x^2 - y^2}$.

$$z = \sqrt{9 - x^2 - y^2}$$

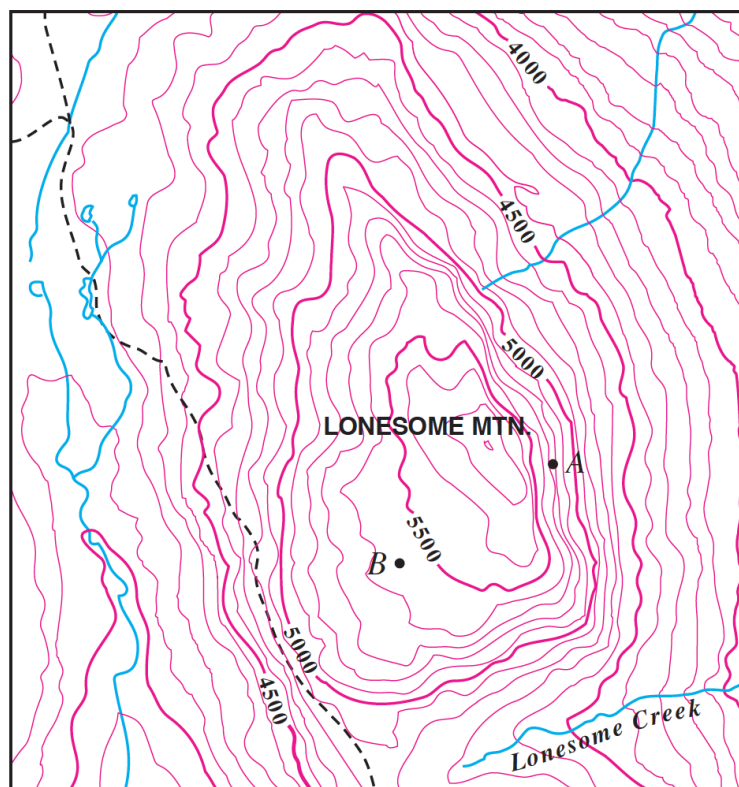
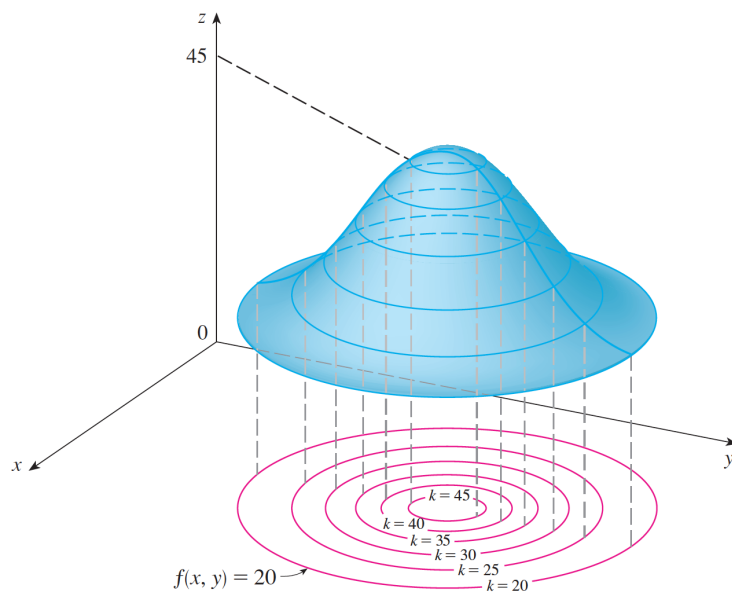
$$\Rightarrow z^2 = 9 - x^2 - y^2$$

$$\Rightarrow x^2 + y^2 + z^2 = 9 \quad \leftarrow \text{a sphere centred at } (0, 0, 0) \text{ with radius } 3$$

Since $z \geq 0$, the graph is the top half of the sphere.



- Level Curves** The level curves of a function f of two variables are the curves with equations $f(x, y) = k$, where k is a constant in the range of f . A collection of such curves is a **contour plot** or a **contour map**.



- **Example** Sketch the level curves of the function $f(x, y) = 6 - 3x - 2y$ for the values $k = -6, 0, 6, 12$.

$$\underline{k = -6}$$

$$-6 = 6 - 3x - 2y$$

$$\Rightarrow 3x + 2y = 12$$

$$\underline{k = 0}$$

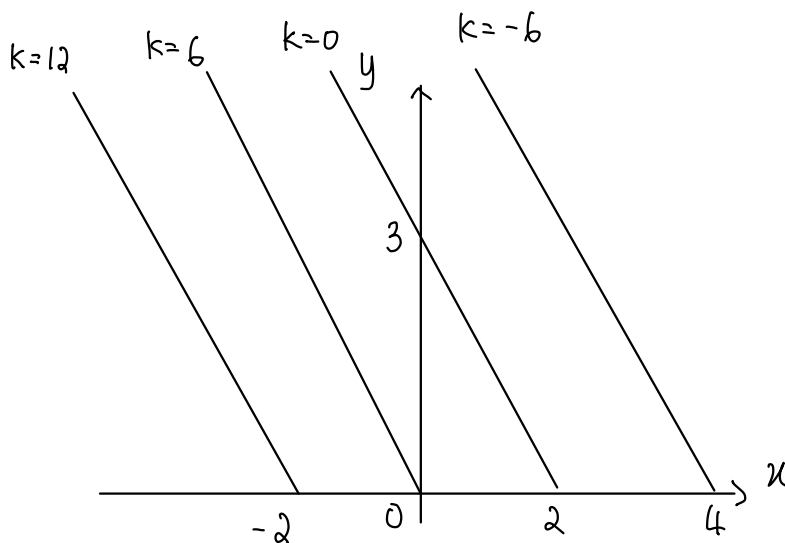
$$3x + 2y = 6$$

$$\underline{k = 6}$$

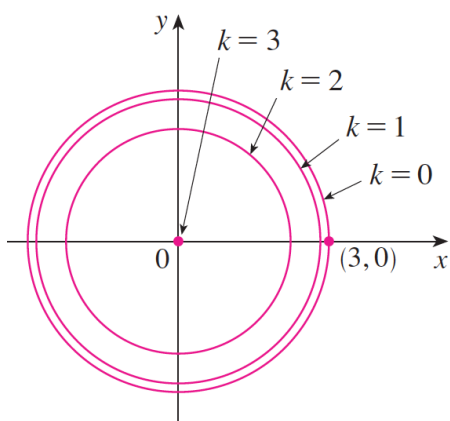
$$3x + 2y = 0$$

$$\underline{k = 12}$$

$$3x + 2y = -6$$



- **Example** Let $g(x, y) = \sqrt{9 - x^2 - y^2}$. Describe the level curves for $k = 0, 1, 2, 3$.



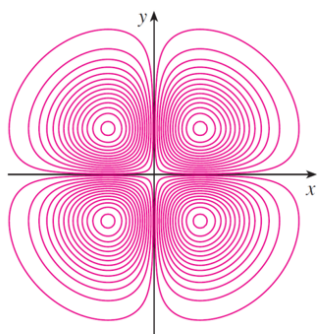
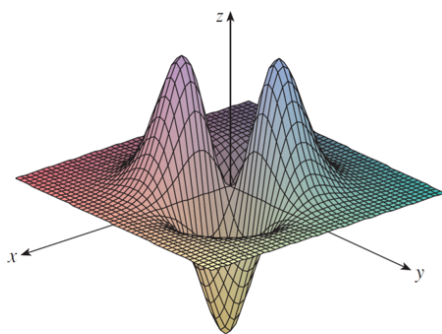
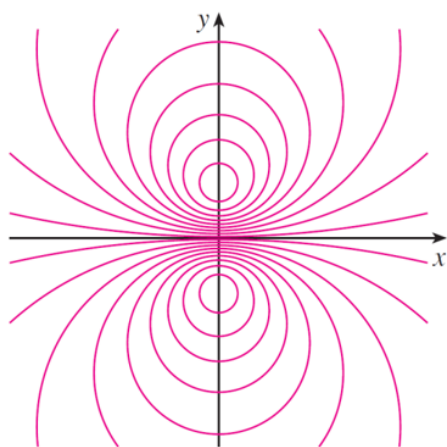
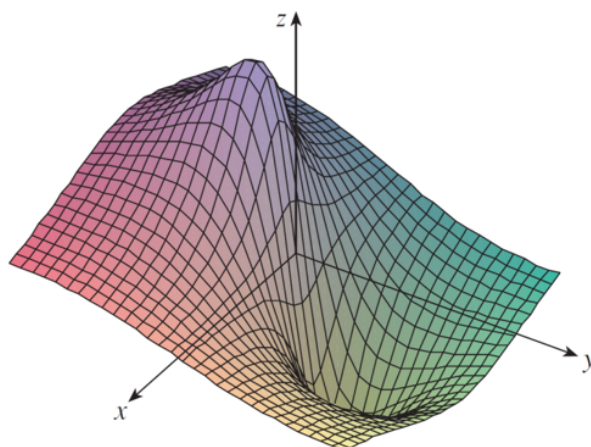
$$\sqrt{9 - x^2 - y^2} = k$$

$$\Rightarrow 9 - x^2 - y^2 = k^2$$

$$\Rightarrow x^2 + y^2 = 9 - k^2$$

level curves are circles centred at $(0, 0)$
with radius $\sqrt{9 - k^2}$.

- **Matching Contour Map and Graph**

(a) Level curves of $f(x, y) = -xye^{-x^2-y^2}$ (b) Two views of $f(x, y) = -xye^{-x^2-y^2}$ (c) Level curves of $f(x, y) = \frac{-3y}{x^2 + y^2 + 1}$ (d) $f(x, y) = \frac{-3y}{x^2 + y^2 + 1}$

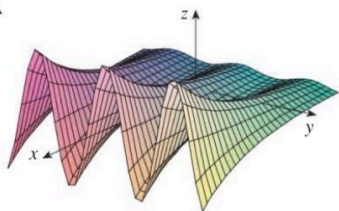
- **Functions of Three or More Variables** In the same way, we can define functions of three or more variables $f(x, y, z)$, $f(x_1, x_2, \dots, x_n)$.

- **Example** Find the domain of $f(x, y, z) = \ln(z - y) + xy \sin z$.

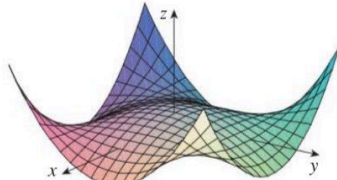
$$\ln(z - y) \Rightarrow z - y > 0 \Rightarrow y < z$$

$$\therefore D = \{(x, y, z) \mid x \in \mathbb{R}, y < z\}$$

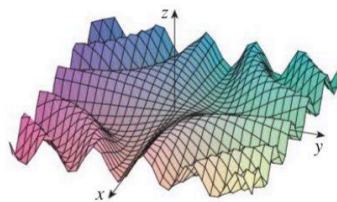
A



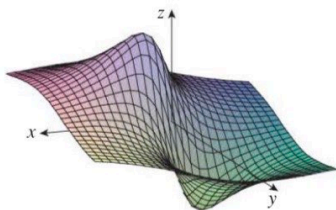
B



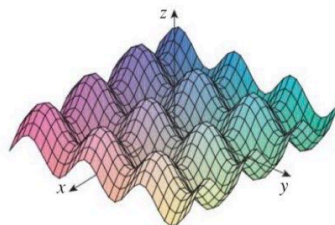
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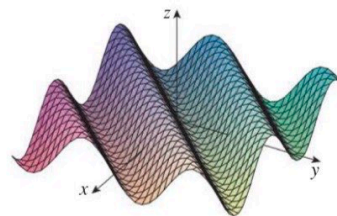
D



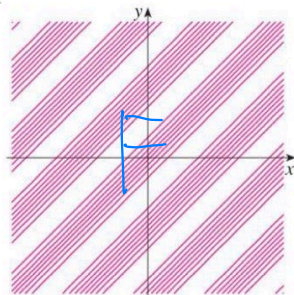
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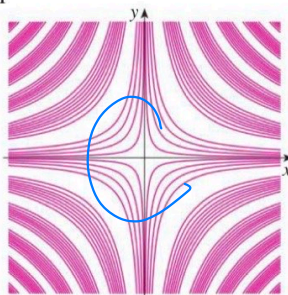
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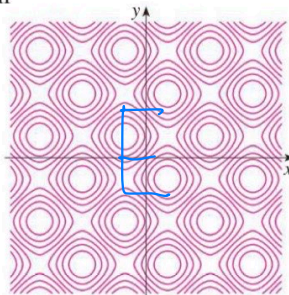
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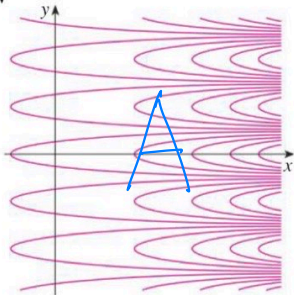
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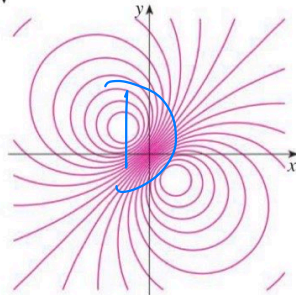
III



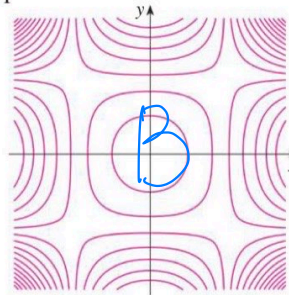
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V



VI



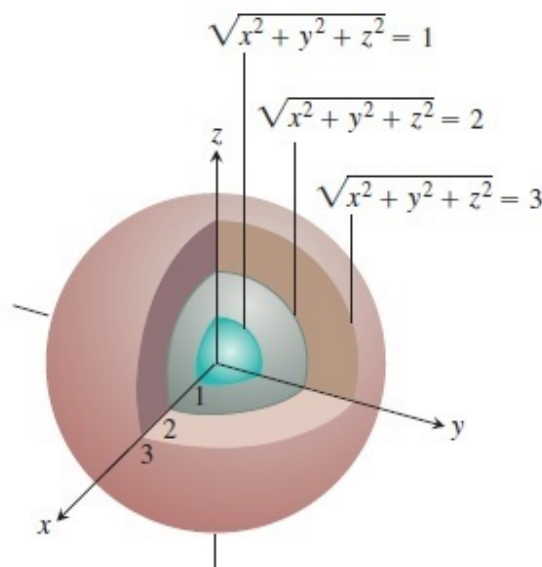
- **Level Surfaces** Given a function of three variables $u = f(x, y, z)$, the surface $\{(x, y, z) : f(x, y, z) = k\}$ in $\mathbb{R}^3$ is called a level surface.
- **Example** Find the level surfaces of the function

$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2}.$$

$$k = \sqrt{x^2 + y^2 + z^2}$$

$$\Rightarrow x^2 + y^2 + z^2 = k^2$$

Level surfaces are spheres centred at $(0, 0, 0)$
with radius k .



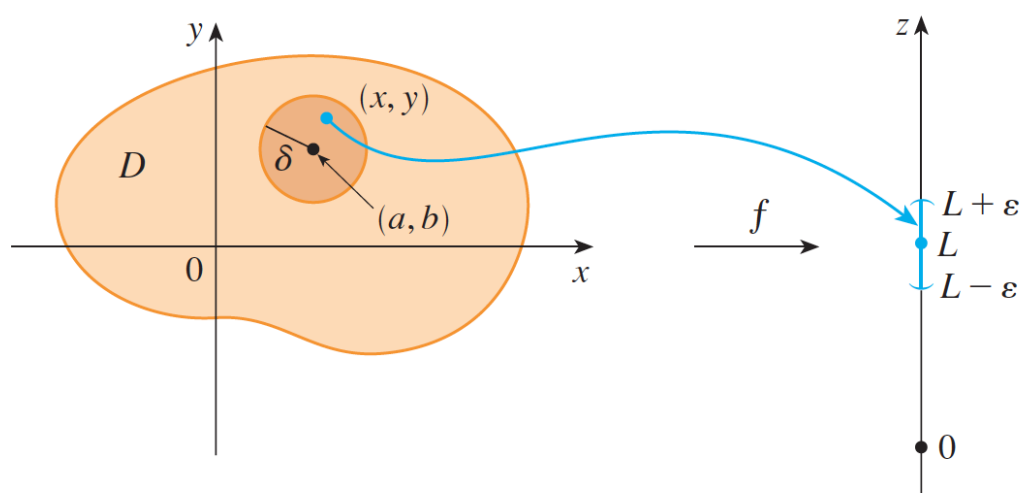
14.2 Limits and Continuity

• Limit of a Function of Two Variables

Let f be a function with domain D that includes points arbitrarily close to (a, b) . Then we say that the **limit** of $f(x, y)$ as (x, y) approaches (a, b) is L , and we write

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L,$$

if for every $\epsilon > 0$, there is a $\delta > 0$ (which may depend on f , ϵ , a , b) such that if $(x, y) \in D$ and $0 < \|(x, y) - (a, b)\| < \delta$, then $|f(x, y) - L| < \epsilon$.

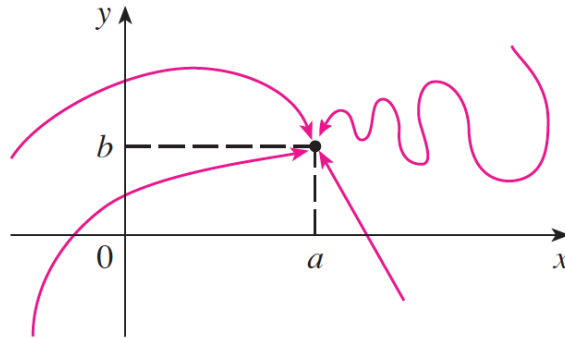


- This definition can be extended to functions of three or more variables.

• Note:

1. $f(a, b)$ need not be defined.

2. If the limit exists, $f(x, y)$ must approach the same limit no matter which path (x, y) takes to approach (a, b) .



If $f(x, y) \rightarrow L_1$ as $(x, y) \rightarrow (a, b)$ along a path C_1 and $f(x, y) \rightarrow L_2$ as $(x, y) \rightarrow (a, b)$ along a path C_2 , where $L_1 \neq L_2$, then $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ **does not exist**.

• Theorem

If $f(x, y)$ is a **polynomial**, then

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$$

If $f(x, y) = \frac{p(x, y)}{q(x, y)}$ is a **rational function** (p, q are polynomials), then

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = \frac{p(a, b)}{q(a, b)} \quad \text{provided } q(a, b) \neq 0$$

If $\lim_{(x,y) \rightarrow (a,b)} p(x, y) = L \neq 0$ and $\lim_{(x,y) \rightarrow (a,b)} q(x, y) = 0$ then

$$\lim_{(x,y) \rightarrow (a,b)} \frac{p(x, y)}{q(x, y)} \text{ does not exist.}$$

- **Example** Guess the following limits based on the numerical evidence:

(a) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} \simeq 1$

$x \backslash y$	-1.0	-0.5	-0.2	0	0.2	0.5	1.0
-1.0	0.455	0.759	0.829	0.841	0.829	0.759	0.455
-0.5	0.759	0.959	0.986	0.990	0.986	0.959	0.759
-0.2	0.829	0.986	0.999	1.000	0.999	0.986	0.829
0	0.841	0.990	1.000		1.000	0.990	0.841
0.2	0.829	0.986	0.999	1.000	0.999	0.986	0.829
0.5	0.759	0.959	0.986	0.990	0.986	0.959	0.759
1.0	0.455	0.759	0.829	0.841	0.829	0.759	0.455

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ does not exist

$x \backslash y$	-1.0	-0.5	-0.2	0	0.2	0.5	1.0
-1.0	0.000	0.600	0.923	1.000	0.923	0.600	0.000
-0.5	-0.600	0.000	0.724	1.000	0.724	0.000	-0.600
-0.2	-0.923	-0.724	0.000	1.000	0.000	-0.724	-0.923
0	-1.000	-1.000	-1.000		-1.000	-1.000	-1.000
0.2	-0.923	-0.724	0.000	1.000	0.000	-0.724	-0.923
0.5	-0.600	0.000	0.724	1.000	0.724	0.000	-0.600
1.0	0.000	0.600	0.923	1.000	0.923	0.600	0.000

- **Example** Evaluate the following limits if they exist:

(a) $\lim_{(x,y) \rightarrow (1,2)} (x^2y + 3y)$ polynomial

$$= (1)^2(2) + 3(2)$$

$$= 8$$

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2 + 1}{x^2 - y^2}$ rational function

$$\lim_{(x,y) \rightarrow (0,0)} x^2 + y^2 + 1 = 0^2 + 0^2 + 1 = 1$$

$$\lim_{(x,y) \rightarrow (0,0)} x^2 - y^2 = 0^2 - 0^2 = 0$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2 + 1}{x^2 - y^2} \text{ does not exist.}$$

- **Example** Show that the function $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$ has no limit at the origin.

Approach $(0, 0)$ along the x -axis, $y = 0$,

$$f(x, 0) = \frac{x^2}{x^2} = 1 \quad \text{for all } x \neq 0.$$

$f(x, y) \rightarrow 1$ as $(x, y) \rightarrow (0, 0)$ along the x -axis.

Approach $(0, 0)$ along the y -axis, $x = 0$,

$$f(0, y) = \frac{-y^2}{y^2} = -1 \quad \text{for all } y \neq 0.$$

$f(x, y) \rightarrow -1$ as $(x, y) \rightarrow (0, 0)$ along the y -axis.

Since f has two different limits along two different paths,

$$\therefore \lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 - y^2}{x^2 + y^2} \text{ does not exist.}$$

- **Example** Evaluate the following limits if they exist:

$$(a) \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{3x^2 + 3y^2}$$

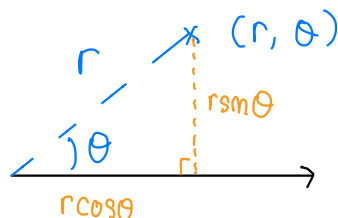
Sometimes, it is easier to work with polar coordinates.

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$\Rightarrow r = \sqrt{x^2 + y^2}$$



$$\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) = 0 \quad \text{so} \quad (x,y) \rightarrow (0,0) \Rightarrow r \rightarrow 0^+$$

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{3x^2 + 3y^2} &= \lim_{r \rightarrow 0} \frac{\sin r^2}{3r^2} \\ &= \frac{1}{3} \left(\lim_{r \rightarrow 0} \frac{\sin r^2}{r^2} \right) \\ &= \frac{1}{3} \end{aligned}$$

$$(b) \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$$

$$= \lim_{r \rightarrow 0} \frac{(r \cos \theta)(r \sin \theta)}{r^2}$$

$$= \lim_{r \rightarrow 0} \cos \theta \sin \theta$$

$$= \cos \theta \sin \theta \quad \leftarrow \theta \text{ can take any value!}$$

Different θ yields different limit :

$$\theta = 0 \quad \Rightarrow \quad \text{limit} = 0$$

$$\theta = \frac{\pi}{4} \quad \Rightarrow \quad \text{limit} = \frac{1}{2}$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} \text{ does not exist.}$$

Another way :

$$\text{Along } x\text{-axis, } y = 0 \Rightarrow f(x, 0) = \frac{0}{x^2} = 0 \text{ for all } x.$$

$$\text{Along } y = x \Rightarrow f(x, x) = \frac{x^2}{x^2 + x^2} = \frac{1}{2}$$

Different limits for different paths, the limit does not exist.